

1 Theoretical Fundation

The exercise consists of determining the position of objects in space, depending on the distance between them.

Considering a matrix:

$$\mathbf{D} = \begin{pmatrix} 0 & d_{12} & \dots & d_{1n} \\ d_{21} & 0 & \dots & d_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ d_{n1} & d_{n2} & \dots & 0 \end{pmatrix} \quad (1)$$

which represents the pairwise distance between the objects. The Matrix is symmetric.

The objective is to determine the position of the objects, in this case 2 dimensional, represented with the matrix:

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \\ \vdots & \vdots \\ x_{n1} & x_{n2} \end{pmatrix}. \quad (2)$$

1.1 Solution

To solve this problem, first a matrix $\mathbf{A}$ is defined as:

$$\mathbf{A} = -\frac{1}{2} \begin{pmatrix} 0 & d_{12}^2 & \dots & d_{1n}^2 \\ d_{21}^2 & 0 & \dots & d_{2n}^2 \\ \vdots & \vdots & \ddots & \vdots \\ d_{n1}^2 & d_{n2}^2 & \dots & 0 \end{pmatrix} \quad (3)$$

this matrix is used to calculate the centred inner-product matrix:

$$\mathbf{B} = \mathbf{H} \mathbf{A} \mathbf{H} \quad (4)$$

where $\mathbf{H}$ is defined as the centering matrix:

$$\mathbf{H} = \mathbf{I}_n - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^T \quad (5)$$